MIL-A-22858A(AS)
AMENDMENT 5
20 July 1973
SUPERSEDING
AMENDMENT 4
31 May 1973

MILITARY SPECIFICATION

ATTITUDE INDICATING SYSTEM, REMOTE STANDBY, TWO INCH

This Amendment forms a part of Military Specification MIL-A-22858A(AS) dated 27 October 1966 and has been approved by the Naval Air Systems Command, Department of the Navy.

Page 1, Under Specifications MILITARY: Delete

"MIL-P-116, MIL-C-4150 and Titles"

Page 2, Under Specifications MILITARY: Add

MIL-C-5541	Chemical Films For Aluminum and Aluminum Alloys
MIL-L-6085	Lubricating Oil, Aircraft Instrument, Low Volatility
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing for Instrument Cover Glasses and Lighting Wedges
MIL-G-23083	Gyroscope, Rate Switching MS17399
MIL-G-81322	Grease, Aircraft, General Purpose Wide Temperature Range
MIL-G-81559	Gyroscopic Assemblies, Attitude and Directional Reference Instruments for Aircraft Preservation and Packaging, Procedures for
MIL-G-81744	Barrier Coating Solution and Solvent

*Page 2, ADD

Naval Air Systems Command

FSC 6610

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AS-2844 Lubricating Oil, Instrument Ball Bearing, High Flash Point

Page 2, Under Standards MILITARY: Delete

"MIL-STD-129 and MIL-STD-794 and Titles" and add
"MIL-STD-471 Maintainability Demonstration"
"MIL-STD-1334 Barrier Coating Process for Anti-Friction Bearings"
and "MS17399 Gyroscope Rate Switching".

Page 3, Paragraph 3.1: Delete and substitute:

"Qualification - This specification makes provision for Qualification Testing (See 6.12)"

Page 4, Add

"3.2.2 Aluminum Alloy Parts - Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

3.2.2.1 Dials, small holes and case inserts need not be anodized.

3.2.2.2 Castings containing aluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.2.3 Non-Magnetic Materials - Non-magnetic materials shall be used for all parts of the indicators except where magnetic materials are essential."

Page 4, Paragraph 3.3.2.3: Delete and substitute

"3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The displacement gyro and indicator shall each have an MTBF of 2000 hours. The rate switching gyro shall have an MTBF in accordance with MIL-G-22083."

* Page 8, Add

"3.3.14 Degree of Cleanliness - The equipment shall be assembled in a Class 100 Clean Room as specified in Fed-Std-209.

3.3.15 Lubrication - The gimbal bearings shall be lubricated with oil AS 2844 or an oil conforming to MIL-L-6085. Teflon separators shall be used in Gimbal Bearings. The spin rotor bearing shall be lubricated with grease MIL-G-81322. The maximum particle size shall not exceed 50 MICRONS. All ball bearings shall be ABEC-5 grade or greater.

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3.3.16 Barrier Coated Bearings -All ball bearings and interfacing associated parts which may bridge and cause migration of lubricant shall be barrier coated with MIL-B-81744 in accordance with MIL-STD-1334. Interfacing parts need not be processed if ball bearings are adequately chamfered with chamfers barrier coated.

3.3.17 Encapsulated Brushes - Encapsulated brush and slip ring assemblies with optimum preset brush tension shall be installed on the outer gimbal axis to prevent contamination and gyro malfunctions. The ball bearings shall be barrier coated where they are provided. Lead wires shall be stranded. The inner gimbal shall also have optimum preset brushes with minimal slip ring runout.

Page 8: Add

"3.4.1.5 Longitudinal Acceleration - The system, when subjected to an 8g acceleration for 4 seconds along the roll axis, shall provide a pitch and roll accuracy of 2 degrees minimum.

3.4.1.6 Coordinated Turns - The system, when subjected to coordinated turns with rates of turn between 15 and 1500 degrees per minute and bank angles to 60 degrees, shall provide a pitch and roll accuracy of 3 degrees minimum after 8 minutes.

3.4.1.7 Uncoordinated Turns - The system, when subjected to uncoordinated turns (skid turns) of less than 5 degrees per minute at air speeds not greater than 400 knots for a 10-minute duration, shall provide a pitch and roll accuracy of 2 degrees minimum."

Page 9, Paragraph 3.5.1.1, Line 2. Delete "Figure 2" and substitute "Figure 2A."

Page 9, Paragraph 3.5.1.2: Delete and substitute:

"3.5.1.2 Shock Mount - The displacement gyro shall have internal vibration isolators. The outline dimensions of either configuration shall conform with Figure 2A."

Page 9, Paragraph 3.5.1.4: Delete "0.4 degree" and substitute "0.30 degree."

Page 10: Delete Figure 2 and substitute Figure 2A of this Amendment.

Page 12: Paragraph 3.5.1.10: Add "The contacts shall also open when wheel speed drops below 5000 RPM."

Page 19; Paragraph 3.5.2.4: First sentence, delete and substitute

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"Means shall be incorporated to adjust the position of the spheroid by means of a pitch trim knob mounted on the lower right side of the face of the indicator. The pitch trim circuit shall be so designed that when the pitch trim knob is placed in zero pitch trim position, there shall be zero volts at the wiper of the pitch trim potentiometer. Zero volts at the pitch trim potentiometer shall cause the voltage between pins A and M to become zero or at null. The servo system shall operate properly for any pitch trim adjustment."

Page 23: Add

"3.5.2.16 Anti-Reflective Coatings - A magnesium fluoride coating in accordance with MIL-C-14806 shall be applied to each surface of the cover glass and the lighting wedge. Coated surfaces shall not chip, scale, peel, flake, dissolve, discolor or otherwise be adversely affected by the environmental tests specified herein or in normal service use. The coating on the outer surface of the glass shall be capable of withstanding abrasion incident to normal service usage."

Page 23 Paragraphs 3.5.3 thru 3.5.3.3: Delete and substitute

"3.5.3 Rate Switching Gyroscope - The rate switching gyroscope shall be in accordance with MIL-G-23083. Dimensions shall be in accordance with MS17399."

Pages 24, 25, and 26, Paragraphs 4.1.1, 4.2, 4.2.1, 4.2.2, and 4.2.3 delete and substitute:

"4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Qualification Tests
(2) Acceptance Tests
(3) Longevity Tests

4.2 Qualification Inspection - The qualification inspection shall consist of all the tests of 4.8 performed in the order listed and shall include the reliability test for the Reliability Qualification Test Phase.

4.2.1 Qualification Inspection Samples: Qualification Inspection Samples shall consist of three units each, manufactured in accordance with this specification. The units submitted for qualification inspection shall have been previously subjected only to the inspections of 4.4.1 and 4.4.2.1. Samples and data shall be forwarded at Suppliers/ Manufacturers expense to the laboratory designated in the Letter of Authorization. Samples shall be retained by the contractor for the Reliability

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Qualification Phase to be run by the contractor at his plant.
4.2.2 Qualification Inspection Sample Identification - The qualification inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information.

> Sample for Qualification Inspection
> (Name of Equipment)
> (Submitted by (Manufacturer's name, date and part number) for Qualification Inspection in accordance with Specification MIL-I-22858(AS) under authorization (reference letter authorizing tests)).

Page 25, Paragraph 4.2.4: Delete "preproduction and substitute "Qualification Inspection."

Pages 25 and 26, Paragraphs 4.3. through 4.3.4: Delete

Page 28, Paragraph 4.4.1: Delete "(dd) (4.8.29)" through "(ff) (4.8.31)"

Add "(ee) Vibration Error (4.8.46)."

Reletter in proper sequence.

Page 28, Paragraph 4.4.2: Delete and substitute

"4.4.2 Sampling Plans - The Sampling Plans shall consist of Sampling Plan A and Sampling Plan B tests. The test samples selected for sampling tests shall first have passed the Individual Test. The test samples which have been subjected to Sampling A test shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual tests. Test samples which have been subjected to the Sampling Plan B test shall not be delivered on contract.

Quantity Offered for Acceptance	Quantity to be Selected for Sampling Plan A Tests
First 15	1 (Zero when Sampling Plan B is invoked)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

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Page 28, Paragraph 4.4.2.1: To title add "Plan A"

In (n) Delete "4.8.45" and substitute
"MIL-G-23083 (4.7.5.)"
Delete "(o) Vibration Error (4.8.46)"

When a defective instrument is detected, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of the next fifteen and proceed as indicated (one from next 50, etc.).

Page 29: Add

4.4.2.2 Sampling Plan B Test - One instrument shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. Unless otherwise directed, these samples shall be forwarded at the contractor's expense to a government laboratory designated by the procuring activity. Each sample shall be plainly identified by a durable tag, securely attached and marked with the following information:

(Name of Equipment)
Submitted by (Manufacturer's name, date)
for Production Acceptance Sampling Plan B
Test of MIL-A-22858 in accordance with
Contract/Order No._____ _____
Manufacturer's Part No._____

4.4.2.2.1 Sampling Plan B Approval - Approval of Sampling Plan B instruments shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material or performance defect made evident during this test shall be corrected by the Contractor to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of instruments under the contract to cease until proper corrective action is approved and accomplished.

4.4.2.2.2 Sampling Plan B Inspection - Each sample selected for Sampling Plan B testing shall be subjected to the following tests:

(a)	Sampling Plan A Tests	
(b)	Vibration Failure	4.8.47
(c)	Radio Noise Interference	4.8.48
(d)	High Temperature Exposure	4.8.49
(e)	High Altitude-Low Temperature	4.8.50

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(f)	Temperature Shock	4.8.51
(g)	Impact Shock	4.8.52
(h)	Acceleration	4.8.53
(j)		
(k)	Humidity	4.8.55
(l)	Salt Spray	4.8.56
(m)	Sand and Dust	4.8.57

Page 29, Paragraph 4.4.3.1: In the first sentence Delete

"Prior to the acceptance of equipment under the contract or order" and substitute "Prior to approval for listing on the Qualified Products List ——."

To the next to last sentence add

", except that temperature extremes shall be +71C and -54C, and the vibration shall be 2 g's on the gyro and 1/2 g on the indicator.

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Page 29, Paragraph 4.4.3.2 and 4.4.3.2.1: Delete and substitute

"4.4.3.2 Reliability Acceptance Phase - Reliability production acceptance testing shall be conducted through an acceptance burn-in. Each equipment produced but not submitted to the qualification test shall be operated for 120 operating hours burn-in, and the test data (operating hours and failures) shall be compared to the accept/reject criteria of Test Plan II of MIL-STD-781. Delivery of equipment shall not be held up to await a decision under the test plan. In event an accept decision is reached, the test duration of each equipment shall be reduced to eighty operating hours. At any time that the current total of test hours and test failures plotted in Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop acceptance of the equipment at any time that a reject situation exists pending a review of the contractors' efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation shall show other than a rejection decision. For the production acceptance phase, test level F shall be used, except that the temperature extremes shall be +55°C and -44°C, and vibration shall be 0.5 g with equipment hard mounted."

Page 32, Paragraph 4.6: Delete in its entirety.

Page 34, Paragraph 4.8.3, Add "The above erection time shall also be met without Scoresby motion and/or vibration."

Page 35, Paragraph 4.8.8: Add "With power on, pins X and Y shall open below a wheel speed of 5000 RPM."

Page 36, Paragraph 4.8.10: Fifth sentence, Delete 0.40 degree, substitute 0.30 degree.

Page 38, Paragraph 4.8.17.1, Third Line: Delete "Pin M" and substitute "Pin N"

Page 39, Make the same change to paragraphs 4.8.18.3 and 4.8.19

Page 41 and 42, Paragraph 4.8.29 through 4.8.32: Delete in entirety.

* Page 43, Paragraph 4.8.33 (a) To sentence add: "Except that normal erection shall be 1° to 3.5° per minute.

Page 43, Paragraph 4.8.34, fourth sentence: Delete and substitute

"At the end of the 10 minute period and while still at the reduced temperature, the equipment shall meet the Individual Tests of 4.4.1 with the following exceptions:"

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Page 43, Paragraph 4.8.34, Subparagraph headed "Rate Switching Gyroscope": Delete in entirety.

Page 44, Paragraph 4.8.35, second sentence: Delete and substitute

"At the end of the four hour period, and while still at the elevated temperatures, the equipment shall meet the individual tests of 4.4.1 with the following exceptions:

Page 44, Paragraph 4.8.35, Subparagraph headed "Rate Switching Gyroscope": Delete in its entirety.

Page 46, Paragraph 4.8.39: Add subparagraph

"4.8.39.1 Fogging - The indicator shall be energized and placed in a +71°C ±2°C controlled ambient temperature for a minimum period of 1 hour. After this period and while at this temperature, an ice cube shall be rubbed on the indicator glass face for a period of 1 minute ± 5 seconds. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog. Evidence of fogging on the inner surface of the cover glass shall be cause for rejection."

Page 47, Paragraph 4.8.45: Delete and substitute

"4.8.45 Operational Accelerations - By flight test by the procuring agency if required.

4.8.45.1 Longitudinal Accelerations - The equipment shall meet the requirements of 3.4.1.5.

4.8.45.2. Coordinated Turns - The equipment shall meet the requirements of 3.4.1.6.

4.8.45.3 Uncoordinated Turns - The equipment shall meet the requirements of 3.4.1.7.

*Page 47, Paragraph 4.8.46 Delete paragraph and substitute

4.8.46 Displacement Gyroscope - The Displacement Gyroscope shall be subjected to a 20 minute scanned vibration resonance test in the longitudinal, lateral and vertical axes. A logarithmic rate of frequency change shall be used from 5 Hz to 500 and back to 5 Hz. Power shall be applied and the case displaced one degree in roll and pitch for resonance search and dwell modes. The vibration level from 5 Hz to 20 Hz shall be 0.1 inch double amplitude. The vibration level from 20 Hz to 500 Hz shall be 2 gs. The signal outputs (pitch and roll) shall be recorded on a strip chart recorder as a permanent record. A five minute dwell

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in each axis, at the most severe resonance frequencies, where the amplitude exceeds 1/2°, shall be made at the applicable vibration level. If more than three significant resonances are found, then the three most severe shall be chosen with at least two of the most severe being selected within the frequency range of 10 Hz to 35 Hz. If the level exceeds 1/2 degree and a predominant resonance cannot be determined then three frequencies separated by at least 7 Hz within the frequency range of 10 Hz to 35 Hz shall be selected for the dwell requirements. Production equipment subjected to the requirements of paragraph 4.3.6. individual tests shall be subjected to a five minute dwell in each axis at one of the most significant resonant frequencies between 10 Hz and 35 Hz determined during Preproduction testing. The pitch and roll outputs shall be monitored 100% of the time. During any five minute dwell failure to meet the following requirements shall be cause for rejection.

(1) The Pitch and Roll outputs shall not change more than + or -0.5 degree from the starting point during the 5 minute dwell in each axis.

(2) There shall be no indication of output signal discontinuities or gimbal binding in either pitch or roll.

(3) The modulation of the output signals shall be less than 15% of the signal at 1° displacement.

(4) The synchro outputs in both axes shall change less than 0.5 degree during the 15 minute vibration period. Change is determined by placing the gyroscope on a table leveled within ± 6 minutes of arc before and after the vibration test.

NOTE: Starting point is defined as the recorded pitch and roll reading taken immediately prior to application of the specified Vibration level, with the gyroscope mounted on the vibrator and the case displaced one degree in roll and pitch.

Page 48, Paragraph 4.8.46, Subparagraph headed "Rate Switching Gyroscope": Delete in its entirety.

Page 48, Paragraph 4.8.47, first sentence: Delete and substitute

"The remote attitude indicator shall be tested in accordance with MIL-T-5422 at the vibration level of Curve III. The displacement gyroscope shall be tested in accordance with MIL-T-5422 at a vibration level of Curve IV. The rate switching gyroscope shall be tested in accordance with the requirements of MIL-G-23083. The above curves shall be extended to 2000 Hz.

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Page 48, Paragraph 4.8.49, last sentence: Delete

"and 4.8.29 - Switching Rate"

Page 49, Paragraph 4.8.50, line 9: Delete

"4.8.29 ..... at conclusion of the turn rates."

Page 49, Paragraph 4.8.54: In first sentence after Displacement Gyro add "—and the Remote Attitude Indicator—"

Page 49: Add

"4.8.55 Humidity - The Remote Attitude Indicating System shall be tested in accordance with the Humidity Test of MIL-T-5422. At the end of this test the equipment shall be subjected to and shall meet the requirements of the Individual Tests."

"4.8.56 Salt Spray - The Remote Attitude Indicator System shall be tested in accordance with the Salt Spray test of MIL-T-5422. At the end of this test the equipment shall be subjected to and shall meet the requirement of the Individual Tests."

"4.8.57 Sand and Dust - The Remote Attitude Indicator System shall be tested in accordance with the Sand and Dust test of MIL-T-5422. At the end of this test the equipment shall be subjected to and shall meet the requirements of the Individual Tests."

Pages 50 and 51, Paragraph 5 through 5.4.1: Delete and substitute

"5. Preparation for Delivery -

5.1 Packaging, Packing, Preservation and Marking - Shall be in accordance with MIL-G-81559."

Page 55: Add

"6.12 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. Suppliers are directed to provisions governing qualification (SD-6) and are urged to arrange to have the products they propose to offer to the Federal Government tested for qualification

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in order to be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products list is the Naval Air Systems Command, Department of the Navy, Washington, D.C. 20361 and information pertaining to qualification of products may be obtained from the activity".

"6.13 Maintainability Demonstration - Maintainability Demonstration shall be in accordance with MIL-STD-471 except as specified herein. If the demonstration has been performed on a previous contract or order, it need not be repeated."

An asterisk preceding a page number denotes a change to be made to the specification which did not appear in AMENDMENT-4.

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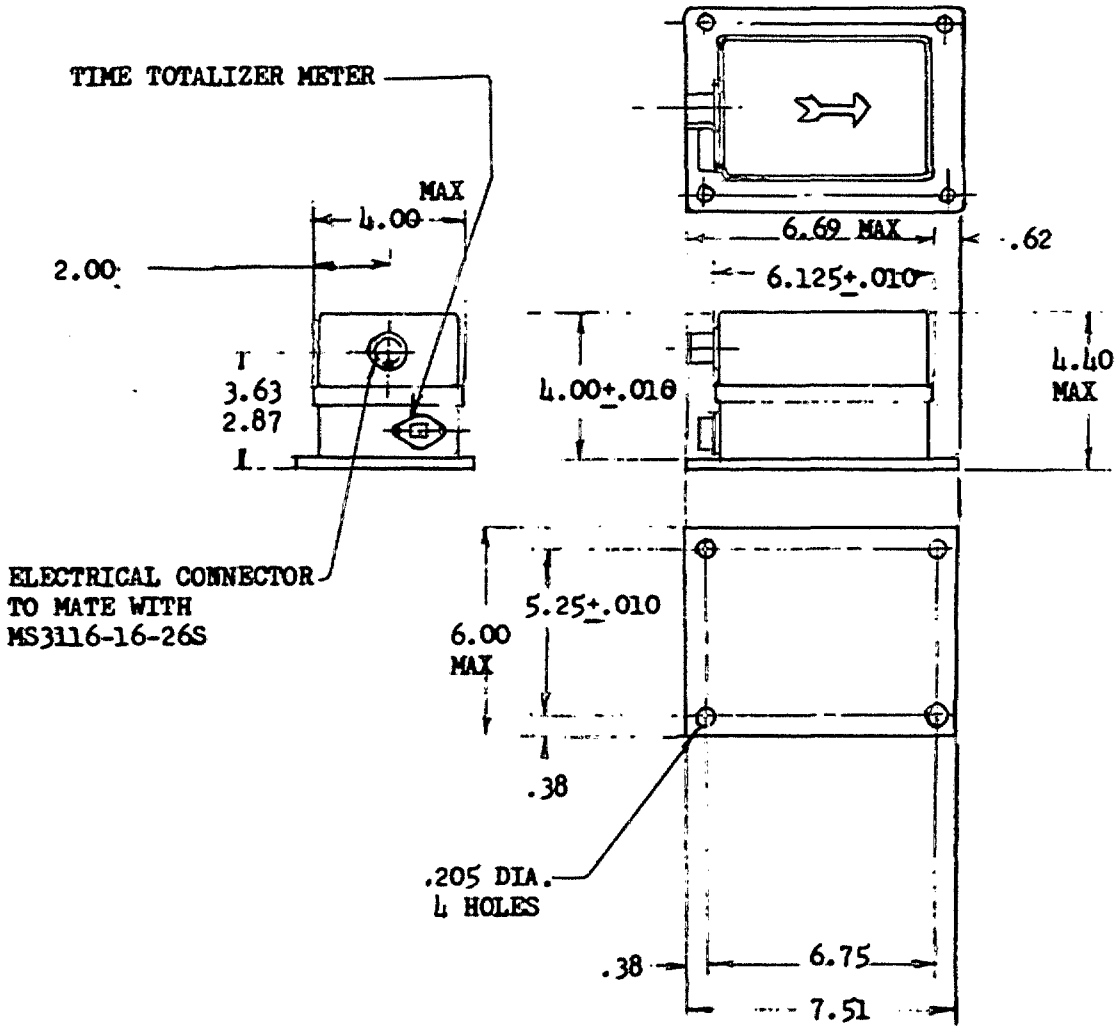


FIGURE 2A OUTLINE DIMENSIONS FOR
DISPLACEMENT GYROSCOPE AND MOUNT ASSEMBLY

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